

Summary

Senior Data Engineer with 10 years of experience designing and operationalizing **scalable data and machine learning solutions** across **consulting, agriculture, and medical industries**. Worked as a **technical lead** on **enterprise data and ML platforms**, owning **architecture decisions** and **end-to-end delivery**. Expertise in **Python, Spark, Kafka, Airflow, Snowflake, Delta Lake, AWS cloud services**, and **production data pipelines**, delivering **reliable, performant, and enterprise-grade analytics and AI solutions**.

Skills & Technologies

- **Data Engineering:** Apache Spark, PySpark, Apache Kafka, Apache Airflow, AWS Glue, Snowflake, Delta Lake, Redshift, Aurora, Hive, Data Lakes & Lakehouses (S3, Lake Formation, Iceberg), ETL/ELT Pipelines (design & technical ownership), Streaming & Real-time Processing (Flink, Kinesis), Big Data Tools (Hadoop, HBase)
 - **Relational Databases:** Postgres, Amazon Aurora, Redshift (expert SQL tuning), working knowledge of Oracle and SQL Server
 - **AI/ML Platforms:** AWS SageMaker (Training Jobs, Endpoints, Pipelines), AWS AI Services (Comprehend, Rekognition, Textract), ML Model Monitoring & Drift Detection (WhyLabs, Evidently AI), Model Governance & Deprecation
 - **Backend & MLOps:** REST/GraphQL APIs, Docker, Kubernetes (EKS), CI/CD (GitHub Actions, Jenkins, ArgoCD), Terraform, Helm, Ansible, AWS Lambda, Step Functions, API Gateway, ECS/Fargate
 - **Security & Governance:** AWS IAM (Roles, Policies), KMS (Key Management), S3 Server-Side Encryption (SSE-S3, SSE-KMS), Data Access Controls
 - **Scripting & OS:** Linux/Unix, Bash shell scripting
 - **Monitoring & Quality:** Prometheus, Grafana, ELK Stack, Datadog, CloudWatch, Data Drift Detection, Pipeline Reliability, Pytest, Great Expectations, ML Observability (Weights & Biases, [Neptune.ai](https://neptune.ai))
-

Professional Experience

McKinsey & Company – Senior Data Engineer with AI/ML (Nov 2023 – Present)

Developed a centralized API management platform for enterprise and healthcare clients, enabling secure integration of internal and client AI/data services. **Worked as a technical lead** on data and API platform initiatives, owning architecture decisions and ensuring production readiness in regulated environments.

- **Worked as a technical lead** on a centralized API management platform, owning system design, delivery execution, and technical standards for enterprise and AI services.
- Built the platform end-to-end, handling backend (Java Spring Boot, Python) and frontend (React/Angular) development.
- Designed and implemented **RESTful and GraphQL APIs**, enabling seamless integration and discovery of client and internal services.
- **Led and supported junior engineers** through design discussions, code reviews, and best-practice adoption.
- Developed automated **ETL and ML data ingestion pipelines** for structured and unstructured datasets supporting AI and analytics workloads.
- Deployed and managed AWS infrastructure (EC2, Lambda, S3, RDS, API Gateway, DynamoDB, Step Functions) using **Terraform** for version-controlled provisioning.
- Implemented **CI/CD pipelines** with Jenkins and GitHub Actions, automating deployments for APIs and ML services.
- Built secure authentication and authorization using **Spring Security and Okta OIDC**, ensuring enterprise-grade compliance.
- Implemented AWS security best practices (IAM, KMS, S3 SSE) to protect sensitive data across data pipelines.
- Applied **DevSecOps practices** (SonarQube, Black Duck) to reduce vulnerabilities and improve code quality.
- **Owned monitoring and reliability** for APIs, ML models, and data pipelines using Prometheus, Grafana, ELK Stack, Datadog, and CloudWatch.
- Integrated **ML model monitoring, drift detection, and governance** using WhyLabs and Evidently AI.

Nike | Senior Data Engineer (Aug 2019 - Nov 2023)

Led data compliance and large-scale analytics projects, building **GDPR/CCPA-compliant pipelines and optimizing Snowflake and Delta Lake** schemas for high-volume retail datasets. Worked closely with the team to provide technical guidance on data platform design and delivery.

- Built a **Self-Delete capability** in the Data Compliance Utility, ensuring **GDPR and CCPA** compliance across user datasets.
- Optimized **Snowflake and Delta Lake** schemas, improving query performance and reducing storage costs for large-scale data pipelines.
- Documented and monitored User Activity Services running on **Airflow and Databricks**, improving observability and operational reliability.
- **Provided technical guidance** during migration of User Permission Services from NSP1 to NSP3, implementing automated **Kafka pipelines** and end-to-end streaming workflows.
- Developed and maintained data pipelines in **Python and SQL**, integrating **Snowflake, Delta Lake, Kafka, Airflow, and Databricks**.
- Developed **PySpark and SparkSQL** jobs for large-scale **ETL** and analytical transformations.
- **Collaborated with cross-functional teams** to execute migration with zero downtime, maintaining seamless service availability.
- Monitored pipeline performance using **Prometheus, Grafana, ELK Stack, and CloudWatch**, ensuring data reliability and governance.
- **Supported and guided junior developers** in building and maintaining scalable data pipelines.

Nutrien | Data Engineer(Feb 2017 - Jul 2019)

Designed and implemented **scalable batch and real-time data pipelines** and a **data lake architecture** to support analytics, operational dashboards, and fraud detection systems. Delivered ETL/ELT workflows, schema optimization, and cloud migration solutions on AWS, improving performance, reducing costs, and enabling enterprise-grade BI and ML workloads.

- Built robust **ETL pipelines** using AWS Glue, PySpark, and Lambda for batch and lightweight transformations, reducing infrastructure costs by 25%.
- Designed and implemented a scalable **data lake** on Amazon S3 with partitioned and compressed (Parquet,ORC,JSON,Avro) formats, improving query performance by 30% via Athena and Redshift Spectrum.
- Developed **real-time analytics pipelines** using Amazon Kinesis, Kafka, and Spark Streaming, achieving sub-5s latency for operational dashboards and fraud detection systems.
- Modeled **dimensional schemas (Star/Snowflake)** in Redshift, enhancing BI performance for Tableau and QuickSight by 3x.
- Orchestrated multi-step workflows using **AWS Step Functions and Glue Workflows**, reducing manual interventions by 40%.
- Migrated **on-prem databases to AWS** using DMS + SCT, automating schema conversion and incremental replication into Aurora and Redshift.
- Developed large-scale **Spark jobs on EMR** for complex data transformations and integrations with Hive and Redshift.
- Worked with self-service data preparation tools and curated datasets to enable analyst-driven transformations
- Monitored **pipeline reliability, data drift, and performance** using Prometheus, Grafana, and CloudWatch.
- **Shared best practices** and supported teammates during development and production support.

Disney |Data Engineer(Jun 2015 - Jan 2017)

Designed and implemented a scalable data discovery portal and backend pipelines for Disney's advertising platforms, enabling efficient navigation of petabytes of data lakes, supporting analytics, reporting, and early ML models for ad optimization. Delivered end-to-end CI/CD pipelines and automated workflows to ensure high reliability, scalability, and maintainability.

- Designed a **data discovery portal from scratch** to navigate petabytes of data lakes, enabling easy access to information and metadata across multiple sources.
- Built **CI/CD pipelines** to deploy the tool via 5 microservices, implementing infrastructure-as-code for scalable, reliable deployments.
- Indexed **Hive metastore** using scheduled and modular **Databuilder DAGs in Airflow**, designed to be extendable to other data warehouses.
- Developed **data ingestion and processing pipelines** to collect, transform, and store high-volume advertising and user interaction data for analytics.

- Collaborated with analytics, ad-ops, and engineering teams to ensure **accurate reporting, dashboarding, and data reliability**.
 - Monitored production pipelines using **Airflow and CloudWatch**, resolving latency, accuracy, and reliability issues.
 - Containerized backend and data services with **Docker**, improving deployment consistency and environment reproducibility.
-

EDUCATION

Bachelor's Degree in Computer Science Engineering (2009 - 2013)
